United States Patent	12413254
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Yu; Changha et al.

Electronic device including antenna

Abstract

An electronic device may include a housing including a first side surface and a second side surface; a support member disposed inside the housing and connected to a part of the first and second side surfaces; a first opening formed between the first side surface and the support member, and a second opening formed among a part of the first side surface, the second side surface, and the support member; a printed circuit board disposed on the support member and having a ground; a first conductive portion disposed between a first segmenting portion formed in the first side surface and a second segmenting portion formed in the second side surface, and including a first ground portion, a first feeding point, a first point, and/or a second point; a second conductive portion disposed between the first segmenting portion and a third segmenting portion formed in the first side surface, and including a second feeding point, a second ground portion, a third feeding point, and/or a third ground portion; a wireless communication circuit electrically connected to the first feeding point, the second feeding point, and/or the third feeding point; a processor electrically connected to the wireless communication circuit; and a first matching circuit electrically connected to the ground and the first point and/or a second matching circuit electrically connected to the ground and the second point, wherein the first conductive portion and at least a part of the second conductive portion may be configured to operate as at least one antenna, depending on an operation of the first matching circuit or the second matching circuit corresponding to control of the processor. Other various embodiments are possible.

Inventors: **Yu; Changha (Suwon-si, KR), Kim; Yunsik (Suwon-si, KR), Kang; Woosuk (Suwon-si, KR), Seo; Mincheol (Suwon-si, KR), Park; Gyubok (Suwon-si, KR), Shin; Donghun (Suwon-si, KR), Lee; Minkyung (Suwon-si, KR)**

Applicant: **SAMSUNG ELECTRONICS CO., LTD.** (Suwon-si, KR)

Family ID: **87520464**

Assignee: **Samsung Electronics Co., Ltd. (Suwon-si, KR)**

Appl. No.: **18/098332**

Filed: **January 18, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230253992 A1	Aug. 10, 2023

Foreign Application Priority Data

KR	10-2022-0016196	Feb. 08, 2022

Related U.S. Application Data

continuation parent-doc WO PCT/KR2022/020297 20221214 PENDING child-doc US 18098332

Publication Classification

Int. Cl.: **H04B1/00** (20060101); **H01Q1/24** (20060101); **H01Q1/48** (20060101); **H01Q9/04** (20060101); **H01Q21/30** (20060101); **H04M1/02** (20060101)

U.S. Cl.:

CPC **H04B1/0064** (20130101); **H01Q1/243** (20130101); **H01Q1/48** (20130101); **H01Q9/045** (20130101); **H01Q21/30** (20130101); **H04M1/0277** (20130101);

Field of Classification Search

CPC: H04B (1/0064); H01Q (1/243); H01Q (1/48); H01Q (9/045); H01Q (21/30); H01Q (5/328); H01Q (5/35); H01Q (1/38); H01Q (1/46); H04M (1/0277); H04M (1/02); H04M (1/026)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11043734	12/2020	Ha	N/A	H01Q 1/48
RE48738	12/2020	Kwak	N/A	H01Q 5/371
11362699	12/2021	Jung	N/A	H04M 1/0202
2014/0210675	12/2013	Hwang	343/702	H01Q 1/44
2015/0123857	12/2014	Park	343/702	H01Q 21/30
2016/0104930	12/2015	Lee	343/702	H01Q 1/243
2017/0201014	12/2016	Lee	N/A	H01Q 1/44
2018/0026337	12/2017	Chen et al.	N/A	N/A
2018/0026361	12/2017	Sakong et al.	N/A	N/A
2018/0288203	12/2017	Jeon et al.	N/A	N/A
2020/0266524	12/2019	Yoon	N/A	G06F 1/1681
2020/0321988	12/2019	Kim et al.	N/A	N/A
2021/0135351	12/2020	Son et al.	N/A	N/A
2021/0391642	12/2020	Yoon et al.	N/A	N/A

2022/0015226	12/2021	Kim et al.	N/A	N/A
2022/0166867	12/2021	Nam et al.	N/A	N/A
2024/0097324	12/2023	Xu	N/A	H01Q 5/35

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
113782974	12/2020	CN	N/A
10-2017-0007086	12/2016	KR	N/A
10-2018-0010957	12/2017	KR	N/A
10-2018-0109509	12/2017	KR	N/A
10-2020-0101234	12/2019	KR	N/A
10-2021-0054262	12/2020	KR	N/A
10-2022-0007947	12/2021	KR	N/A
WO 2019/218167	12/2018	WO	N/A

OTHER PUBLICATIONS

International Search Report dated Mar. 21, 2023 for KR Application No. KR/2022/020297. cited by applicant
Notification of Publication dated Aug. 17, 2023 for KR Application No. KR/2022/020297. cited by applicant
Extended European Search Report dated Mar. 26, 2025 for EP Application No. 22926213.4. cited by applicant

---

Primary Examiner: Sivji; Nizar N

Attorney, Agent or Firm: Nixon & Vanderhye P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/KR2022/020297, filed on Dec. 14, 2022, designating the United States, in the Korean Intellectual Property Receiving Office, and claiming priority to KR Patent Application No. 10-2022-0016196 filed on Feb. 8, 2022, the disclosures of which are all hereby incorporated by reference herein in their entireties.

BACKGROUND
Field
(1) Various example embodiments relate to an electronic device including at least one antenna.
Description of Related Art
(2) The use of an electronic device such as a smart phone of bar type, foldable type, rollable type or sliding type, or a tablet PC is increasing, and various functions are provided to the electronic device.
(3) The electronic device may transmit and receive a phone call and various data with another electronic device through wireless communication.
(4) In order to perform wireless communication with another electronic device using a network, the electronic device may include at least one antenna.
SUMMARY
(5) At least a portion of a housing forming the exterior of the electronic device may be formed of a conductive material (e.g., metal).
(6) At least the portion of the housing formed of the conductive material may be used as an antenna

(e.g., antenna radiator) for performing wireless communication. For example, the housing of the electronic device may be divided through at least one segmenting portion (e.g., a slit) and be used as at least one antenna.

(7) In order to provide various services, the electronic device may additionally form a segmenting portion in the housing and thereby increase the number of antennas. Increasing the number of segmenting portions in the housing of the electronic device may decrease competitiveness in design of the electronic device and strength against external impacts.

(8) The electronic device may include a laser direct structuring (LDS) antenna patterned inside the housing. The LDS antenna may have a limited area to be mounted inside the housing and may reduce a space for arranging electronic components inside the housing.

(9) Various example embodiments may provide an electronic device capable of supporting a variety of frequency bands by allowing a plurality of antennas (e.g., conductive portions) to use one segmenting portion formed in a housing.

(10) The technical problems to be achieved are not limited to the above-mentioned problems, and other technical problems not mentioned are clearly understood from the following description by a person skilled in the art to which the disclosure belongs.

(11) According to various example embodiments, an electronic device may include a housing including a first side surface and a second side surface; a support member disposed inside the housing and connected, directly or indirectly, to a part of the first and second side surfaces; a first opening formed between the first side surface and the support member, and a second opening formed among a part of the first side surface, the second side surface, and the support member; a printed circuit board disposed on the support member and having a ground; a first conductive portion disposed between a first segmenting portion formed in the first side surface and a second segmenting portion formed in the second side surface, and including a first ground portion, a first feeding point, a first point, and/or a second point; a second conductive portion disposed between the first segmenting portion and a third segmenting portion formed in the first side surface, and including a second feeding point, a second ground portion, a third feeding point, and/or a third ground portion; a wireless communication circuit electrically connected, directly or indirectly, to the first feeding point, the second feeding point, and/or the third feeding point; a processor electrically connected, directly or indirectly, to the wireless communication circuit; and a first matching circuit electrically connected, directly or indirectly, to the ground and the first point and/or a second matching circuit electrically connected, directly or indirectly, to the ground and the second point, wherein the first conductive portion and at least a part of the second conductive portion may be configured to operate as at least one antenna, depending on an operation of the first matching circuit or the second matching circuit corresponding to control of the processor.

(12) According to various example embodiments, an electronic device may include a housing including a first side surface and a second side surface; a support member disposed inside the housing and connected to a part of the first and second side surfaces; a first opening formed between the first side surface and the support member, and a second opening formed among a part of the first side surface, the second side surface, and the support member; a printed circuit board disposed on the support member and having a ground; a first conductive portion disposed between a first segmenting portion formed in the first side surface and a second segmenting portion formed in the second side surface, and including a first ground portion, a first feeding point, a first point, and/or a second point; a second conductive portion disposed between the first segmenting portion and a third segmenting portion formed in the first side surface, and including a second feeding point, a second ground portion, a third feeding point, and/or a third ground portion; a wireless communication circuit electrically connected to the first feeding point, the second feeding point, and/or the third feeding point; a processor electrically connected to the wireless communication circuit; and a first matching circuit electrically connected to the ground and the first point and/or a second matching circuit electrically connected to the ground and the second point, wherein

depending on an operation of the first matching circuit or the second matching circuit corresponding to control of the processor, the first conductive portion electrically connected to the first and second matching circuits and a part included in the second conductive portion and corresponding to the second ground portion may be configured to operate as a first antenna, and wherein a part from the second ground portion included in the second conductive portion to the second point included in the first conductive portion and electrically connected to the second matching circuit may be configured to operate as a second antenna.
(13) According to various example embodiments, because a plurality of antennas share and use one segmenting portion formed in a housing, it is possible to reduce the number of segmenting portions formed in the housing, improve design competitiveness, and secure strength against external impacts.
(14) In addition, various effects will be explicitly or implicitly provided in the disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) In connection with the description of the drawings, the same or similar reference numerals may be used for the same or similar components. The above and other aspects, features, and advantages of certain example embodiments will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a block diagram illustrating an electronic device in a network environment according to various example embodiments.
(3) FIG. **2**A is a perspective view illustrating a front surface of an electronic device according to various example embodiments.
(4) FIG. **2**B is a perspective view illustrating a rear surface of an electronic device according to various example embodiments.
(5) FIG. **3** is an exploded perspective view of an electronic device according to various example embodiments.
(6) FIG. **4**A is a diagram schematically illustrating a part of an electronic device including at least one segmenting portion according to various example embodiments.
(7) FIG. **4**B is a schematic enlarged view of the part A shown in FIG. **4**A according to various example embodiments.
(8) FIG. **4**C is a diagram schematically illustrating the circuit configuration of the first conductive portion and the second conductive portion of FIGS. **4**A and **4**B.
(9) FIG. **5** is a diagram schematically illustrating the configurations of the first matching circuit and the second matching circuit shown in FIG. **4**C according to various example embodiments.
(10) FIG. **6**A is a diagram illustrating an operation principle of using, as a first antenna, a first conductive portion and a part of a second conductive portion of an electronic device according to various example embodiments.
(11) FIG. **6**B is a diagram illustrating a flow of current applied to a first conductive portion and a part of a second conductive portion according to various example embodiments.
(12) FIG. **6**C is a diagram illustrating electric fields for a first conductive portion and a part of a second conductive portion according to various example embodiments.
(13) FIG. **7**A is a diagram illustrating an operation principle of using, as a second antenna, an end of a first conductive portion and a part of a second conductive portion of an electronic device according to various example embodiments.
(14) FIG. **7**B is a diagram illustrating a flow of current applied to a part of a first conductive portion and a part of a second conductive portion according to various example embodiments.
(15) FIG. **7**C is a diagram illustrating electric fields for a part of a first conductive portion and a

part of a second conductive portion according to various example embodiments.
(16) FIG. **8**A is a diagram illustrating an operation principle of using, as a third antenna, a part of a first conductive portion and at least a part of a second conductive portion of an electronic device according to various example embodiments.
(17) FIG. **8**B is a diagram illustrating a flow of current applied to a part of a first conductive portion and at least a part of a second conductive portion according to various example embodiments.
(18) FIG. **8**C is a diagram illustrating a radiation pattern of a third antenna of an electronic device according to various example embodiments.
(19) FIG. **8**D is a diagram illustrating radiation efficiency for a third antenna of an electronic device according to various example embodiments and an antenna of an electronic device according to a comparative embodiment.
(20) FIG. **9** is a diagram schematically illustrating various configurations of a first conductive portion and a second conductive portion of an electronic device according to various example embodiments.

DETAILED DESCRIPTION

(21) FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments.
(22) Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).
(23) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.
(24) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication

module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.
(25) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.
(26) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.
(27) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).
(28) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.
(29) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.
(30) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.
(31) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(32) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.
(33) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).
(34) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.
(35) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.
(36) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).
(37) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.
(38) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.
(39) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band)

to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(40) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(41) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(42) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(43) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing

or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(44) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(45) It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via at least a third element.

(46) As used in connection with various example embodiments, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(47) FIG. **2**A is a front perspective view of an electronic device according to various example embodiments. FIG. **2**B is a rear perspective view of the electronic device according to various example embodiments.

(48) Referring to FIG. **2**A and FIG. **2**B, an electronic device **200** according to an embodiment may include a housing **210** including a first surface (or front surface) **210**A, a second surface (or rear surface) **210**B, and a side surface **210**C surrounding the space between the first surface **210**A and the second surface **210**B. In another embodiment (not illustrated), the housing may denote a structure that forms a part of the first surface **210**A, the second surface **210**B, and the side surface **210**C illustrated in FIG. **2**A and FIG. **2**B. According to an embodiment, the first surface **210**A may be formed by a front plate **202**, at least a part of which is substantially transparent (for example, a glass plate including various coating layers, or a polymer plate). The second surface **210**B may be formed by a rear plate **211** that is substantially opaque. The rear plate **211** may be made of coated or colored glass, ceramic, polymer, metal (for example, aluminum, stainless steel (STS), or magnesium), or a combination of at least two of the above-mentioned materials. The side surface **210**C may be formed by a side bezel structure (or “side member”) **218** which is coupled to the front plate **202** and to the rear plate **211**, and which includes metal and/or polymer. In some embodiments, the rear plate **211** and the side bezel structure **218** may be formed integrally and may

include the same material (for example, a metal material such as aluminum).
(49) In the illustrated embodiment, the front plate **202** may include two first areas **210**D on both ends of the long edge of the front plate **202** such that the two first areas **210**D bend from the first surface **210**A toward the rear plate **211** and extend seamlessly. In the illustrated embodiment (see FIG. **2**B), the rear plate **211** may include two second areas **210**E on both ends of the long edge such that the two second areas **210**E bend from the second surface **210**B toward the front plate **202** and extend seamlessly. In some embodiments, the front plate **202** (or the rear plate **211**) may include only one of the first areas **210**D (or the second areas **210**E). In another embodiment, a part of the first areas **210**D or the second areas **210**E may not be included. In the above embodiments, when seen from the side surface of the electronic device **200**, the side bezel structure **218** may have a first thickness (or width) on a part of the side surface, which does not include the first areas **210**D or the second areas **210**E as described above, and may have a second thickness that is smaller than the first thickness on a part of the side surface, which includes the first areas **210**D or the second areas **210**E.
(50) According to an embodiment, the electronic device **200** may include at least one of a display **201**, audio modules **203**, **207**, and **214**, sensor modules **204** and **219**, camera modules **205**, **212**, and **213**, a key input device **217**, indicator, and connector holes **208** and **209**. In some embodiments, at least one of the constituent elements (for example, the key input device **217** or indicator) of the electronic device **200** may be omitted, or the electronic device **200** may additionally include another constituent element.
(51) The display **201** may be exposed through a corresponding part of the front plate **202**, for example. In some embodiments, at least a part of the display **201** may be exposed through the front plate **202** that forms the first areas **210**D of the side surface **210**C and the first surface **210**A. In some embodiments, the display **201** may have a corner formed in substantially the same shape as that of the adjacent outer periphery of the front plate **202**. In another embodiment (not illustrated), in order to increase the area of exposure of the display **201**, the interval between the outer periphery of the display **201** and the outer periphery of the front plate **202** may be formed to be substantially identical.
(52) The audio modules may include a microphone hole **203** and speaker holes **207** and **214**. A microphone for acquiring an external sound may be arranged in the microphone hole **203**, and a plurality of microphones may be arranged therein such that the direction of a sound can be sensed in some embodiments. The speaker holes **207** and **214** may include an outer speaker hole **207** and a speech receiver hole **214**. In some embodiments, the speaker holes **207** and **214** and the microphone hole **203** may be implemented as a single hole, or a speaker may be included (for example, a piezoelectric speaker) without the speaker holes **207** and **214**.
(53) The sensor modules **204** and **219** may generate an electric signal or a data value corresponding to the internal operating condition of the electronic device **200** or the external environment condition thereof. The sensor modules **204** and **219** may include, for example, a first sensor module **204** (for example, a proximity sensor) arranged on the first surface **210**A of the housing **210**, and/or a second sensor module (not illustrated) (for example, a fingerprint sensor), and/or a third sensor module **219** (for example, an HRM sensor) arranged on the second surface **210**B of the housing **210**, and/or a fourth sensor module (for example, a fingerprint sensor). The fingerprint sensor may be arranged not only on the first surface **210**A (for example, the display **201**) of the housing **210**, but also on the second surface **210**B thereof. The electronic device **200** may further include a sensor module not illustrated, for example, at least one of a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or a luminance sensor **204**.
(54) The camera modules **205**, **212**, and **213** may include a first camera device **205** arranged on the first surface **210**A of the electronic device **200**, a second camera device **212** arranged on the second

surface **210**B thereof, and/or a flash **213**. The camera devices **205** and **212** may include a single lens or a plurality of lenses, an image sensor, and/or an image signal processor. The flash **213** may include, for example, a light-emitting diode or a xenon lamp. In some embodiments, two or more lenses (an infrared camera, a wide-angle lens, and a telephoto lens) and image sensors may be arranged on a single surface of the electronic device **200**.
(55) The key input device **217** may be arranged on the side surface **210**C of the housing **210**. In another embodiment, the electronic device **200** may not include a part of the above-mentioned key input device **217** or the entire key input device **217**, and the key input device **217** (not included) may be implemented in another type, such as a soft key, on the display **201**. In some embodiments, the key input device may include a sensor module **216** arranged on the second surface **210**B of the housing **210**.
(56) The indicator may be arranged on the first surface **210**A of the housing **210**, for example. The indicator may provide information regarding the condition of the electronic device **200** in a light type, for example. In another embodiment, the indicator may provide a light source that interworks with operation of the camera module **205**, for example. The indicator may include, for example, an LED, an IR LED, and a xenon lamp.
(57) The connector holes **208** and **209** may include a first connector hole **208** capable of containing a connector (for example, a USB connector) for transmitting/receiving power and/or data to/from an external electronic device, and/or a second connector hole (for example, an earphone jack) **209** capable of containing a connector for transmitting/receiving an audio signal to/from the external electronic device.
(58) In another embodiment (not illustrated), a recess or an opening may be formed in a part of the screen display area of the display **201**, and at least one of an audio module **214**, a sensor module **204**, a camera module **205**, and a light-emitting element **206** may be included and aligned with the recess or the opening. In another embodiment (not illustrated), on the back surface of the screen display area of the display **201**, at least one of an audio module **214**, a sensor module **204**, a camera module **205**, a fingerprint sensor **216**, and a light-emitting element **206** may be included. In another embodiment (not illustrated), the display **201** may be coupled to or arranged adjacent to a touch sensing circuit, a pressure sensor capable of measuring the intensity (pressure) of a touch, and/or a digitizer that detects a magnetic field-type stylus pen. In some embodiments, at least a part of the sensor modules **204** and **219** and/or at least a part of the key input device **217** may be arranged in the first areas **210**D and/or the second areas **210**E.
(59) FIG. **3** is an exploded perspective view of the electronic device according to various example embodiments.
(60) Referring to FIG. **3**, the electronic device **300** may include a side bezel structure **310**, a first support member **311** (for example, a bracket), a front plate **320**, a display **330**, a printed circuit board **340**, a battery **350**, a second support member **360** (for example, a rear case), an antenna **370**, and a rear plate **380**. In some embodiments, at least one of the constituent elements (for example, the first support member **311** or the second support member **360**) of the electronic device **300** may be omitted, or the electronic device **300** may further include another constituent element. At least one of the constituent elements of the electronic device **300** may be identical or similar to at least one of the constituent elements of the electronic device **101** or **200** of FIG. **1** to FIG. **2**B, and repeated descriptions thereof will be omitted herein.
(61) The first support member **311** may be arranged inside the electronic device **300** and connected to the side bezel structure **310**, or may be formed integrally with the side bezel structure **310**. The first support member **311** may be made of a metal material and/or a nonmetal (for example, polymer) material, for example. The display **330** may be coupled to one surface of the first support member **311**, and the printed circuit board **340** may be coupled to the other surface thereof.
(62) A processor, a memory, and/or an interface may be mounted on the printed circuit board **340**. The processor may include, for example, one or more of a central processing device, an application

processor, a graphic processing device, an image signal processor, a sensor hub processor, or a communication processor.

(63) According to various embodiments, the printed circuit board (PCB) **340** may include a first PCB **340***a* and/or a second PCB **340***b*. For example, the first PCB **340***a* and the second PCB **340***b* may be disposed to be spaced apart from each other and electrically connected using a connection member **345** (e.g., a coaxial cable and/or FPCB). In another example, the PCB **340** may include a structure in which a plurality of PCBs are stacked. The PCB **340** may include an interposer structure. The PCB **340** may be implemented in the form of a flexible PCB (FPCB) and/or a rigid PCB. In an embodiment, the side member **310** (e.g., the housing) may include at least one segmenting portion (e.g., a slit).

(64) The memory may include a volatile memory or a non-volatile memory, for example.

(65) The interface may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. The interface may connect the electronic device **300** with an external electronic device electrically or physically, for example, and may include a USB connector, an SD card/MMC connector, or an audio connector.

(66) The battery **350** is a device for supplying power to at least one constituent element of the electronic device **300**, and may include a non-rechargeable primary cell, a rechargeable secondary cell, or a fuel cell, for example. At least a part of the battery **350** may be arranged on substantially the same plane with the printed circuit board **340**, for example. The battery **350** may be arranged integrally inside the electronic device **300**, or may be arranged such that the same can be attached to/detached from the electronic device **300**.

(67) The antenna **370** may be arranged between the rear plate **380** and the battery **350**. The antenna **370** may include, for example, a near field communication (NFC) antenna, a wireless charging antenna, and/or a magnetic secure transmission (MST) antenna. The antenna **370** may conduct near-field communication with an external device or may wirelessly transmit/receive power necessary for charging, for example. In another embodiment, an antenna structure may be formed by a part or a combination of the side bezel structure **310** and/or the first support member **311**.

(68) FIG. **4**A is a diagram schematically illustrating a part of an electronic device including at least one segmenting portion according to various example embodiments. FIG. **4**B is a schematic enlarged view of the part A shown in FIG. **4**A according to various example embodiments. FIG. **4**C is a diagram schematically illustrating the circuit configuration of the first conductive portion and the second conductive portion of FIGS. **4**A and **4**B.

(69) In an embodiment, FIG. **4**A is a diagram, viewed from the negative z-axis direction, of a part (e.g., in the y-axis direction) of a state in which the PCB **340** (e.g., the second PCB **340***a*) is disposed on one surface (e.g., in the negative z-axis direction) of the support member **311** (e.g., the first support member **311**) disposed in the housing **310** (e.g., the side member) shown in FIG. **3** according to various example embodiments.

(70) According to various embodiments, the electronic device **300** disclosed below may include the embodiments of the electronic devices **101** and **200** illustrated in FIGS. **1** to **3**. In the description of the electronic device **300** disclosed below, the same reference numbers may be assigned to the components substantially the same as those of the embodiments illustrated in FIGS. **1** to **3**, and duplicate descriptions of their functions may be omitted.

(71) According to an embodiment, although the embodiment related to the electronic device **300** disclosed below will be described with respect to a bar-type electronic device, which is not a limitation. The following description may also be applied to any electronic device such as a foldable type, a rollable type, a sliding type, a wearable type, a tablet PC, and/or a notebook PC.

(72) With reference to FIGS. **4**A to **4**C, the housing **310** of the electronic device **300** according to an example embodiment may have a first side surface **410**, a second side surface **420**, and a third side surface **430**. The first side surface **410** may extend in the lower direction (e.g., in the negative

y-axis direction) from a first part (e.g., in the negative x-axis direction) of the second side surface **420** disposed in the upper direction (e.g., in the y-axis direction) of the electronic device **300**, and the third side surface **430** may extend in the lower direction (e.g., in the negative y-axis direction) from a second part (e.g., in the x-axis direction) of the second side surface **420**.
(73) According to an embodiment, the housing **310** may include a support member **311** (e.g., the first support member **311** in FIG. **3**) disposed therein. The support member **311** may be connected, directly or indirectly, to the housing **310** or integrally formed inside the housing **310**. For example, the support member **311** may be connected, directly or indirectly, to a part of the first side surface **410**, a part of the second side surface **420**, and a part of the third side surface **430**. Between a part of the first side surface **410** and the support member **311**, a first opening **406** and a part of a second opening **408** may be formed. Between a part of the second side surface **420** and the support member **311**, the second opening **408** may be formed.
(74) According to an embodiment, the first side surface **410** may have a first segmenting portion **401** (e.g., a slit) and/or a third segmenting portion **403**. The first segmenting portion **401** may be formed closer to the first part (e.g., in the negative x-axis direction) of the second side surface than the third segmenting portion **403**. The second side surface **420** may have a second segmenting portion **402**. Between the first segmenting portion **401** and the second segmenting portion **402**, a first conductive portion **415** (e.g., a first radiator) may be disposed. For example, the first conductive portion **415** may have an "¬" shape between the first segmenting portion **401** formed in the first side surface **410** and the second segmenting portion **402** formed in the second side surface **420**.
(75) According to an embodiment, the first conductive portion **415** may have inwardly a first ground portion G**1**, a first feeding point F**1**, a first point M**1** and/or a second point M**2**. For example, the first ground portion G**1** may be a portion that protrudes from the inner side of the first conductive portion **415** in a lower direction (e.g., the negative y-axis direction) and is electrically connected to a part of the support member **311**. The first feeding point F**1** and the first point M**1** may be portions protruding from the inner side of the first conductive portion **415** toward the second opening **408** in a lower direction (e.g., in the negative y-axis direction). The second point M**2** may be a portion protruding from the inner side of the first conductive portion **415** toward the second opening **408** in one direction (e.g., the x-axis direction).
(76) According to an embodiment, the first ground portion G**1** may be disposed adjacent to the second segmenting portion **402**. For example, the first ground portion G**1** may be disposed adjacent to an end, in the x-axis direction, of the first conductive portion **415** adjacent to the second segmenting portion **402**. In another example, the first ground portion G**1** may be positioned between the second segmenting portion **402** and the first feeding point F**1**. The second point M**2** may be disposed adjacent to the first segmenting portion **401**. For example, the second point M**2** may be disposed adjacent to an end, in the negative x-axis direction and the negative y-axis direction, of the first conductive portion **415** adjacent to the first segmenting portion **401**. In still another example, the second point M**2** may be positioned between the first segmenting portion **401** and the first point M**1**. Between the first ground portion G**1** and the second point M**2**, the first feeding point F**1** and the first point M**1** may be disposed. For example, the first feeding point F**1** may be disposed between the first ground portion G**1** and the first point M**1**. The first point M**1** may be disposed between the first feeding point F**1** and the second point M**2**.
(77) According to an embodiment, the first ground portion G**1** may be connected to a part of the support member **311**. The first ground portion G**1** may be electrically connected to the ground G formed on the PCB **340** using a part of the support member **311**. The first ground portion G**1** may ground the first conductive portion **415**.
(78) According to an embodiment, the first feeding point F**1** may be electrically connected to a wireless communication circuit **460** (e.g., the wireless communication module **192** in FIG. **1**, comprising communication circuitry) disposed on the PCB **340** through a first signal path **461**, such

as in FIG. **4**C. The first conductive portion **415** (e.g., a first radiator) may be electrically connected to the wireless communication circuit **460** through the first feeding point F**1**, thus operating as an antenna.
(79) According to an embodiment, the first point M**1** may be electrically connected to a first matching circuit **451** disposed on the PCB **340** through a fourth signal path **454**. The second point M**2** may be electrically connected to a second matching circuit **452** disposed on the PCB **340** through a fifth signal path **455**.
(80) According to an embodiment, the first matching circuit **451** and/or the second matching circuit **452** may be electrically connected to and controlled by a processor **450** disposed on the PCB **340**. Each "processor" herein comprises processing circuitry.
(81) According to an embodiment, between the first segmenting portion **401** and the third segmenting portion **403** formed on the first side surface **410**, a second conductive portion **425** (e.g., a second radiator) may be disposed.
(82) According to an embodiment, the second conductive portion **425** may have inwardly a second feeding point F**2**, a second ground portion G**2**, a third feeding point F**3**, and/or a third ground portion G**3**. For example, the second feeding point F**2** may be a portion protruding from the inner side of the first side surface **410** toward the second opening **408** in the x-axis direction. The second ground portion G**2** may be a portion that protrudes from the inner side of the first side surface **410** in the x-axis direction and is electrically connected to a part of the support member **311**. The third feeding point F**3** may be a portion protruding from the inner side of the first side surface **410** toward the first opening **406** in the x-axis direction. The third feeding point F**3** is formed to be surrounded by the first opening **406** and may be formed between the second feeding part G**2** and the third feeding part G**3**. The third ground portion G**3** may be a portion that protrudes from the inner side of the first side surface **410** in the x-axis direction and is electrically connected to a part of the support member **311**.
(83) According to an embodiment, the second feeding point F**2** may be disposed adjacent to the first segmenting portion **401** in the second opening **408**. For example, the second feeding point F**2** may be disposed adjacent to an end, in the y-axis direction, of the second conductive portion **425** adjacent to the first segmenting portion **401**. In another example, the second feeding point F**2** may be positioned between the first segmenting portion **401** and the second ground portion G**2**. The third ground portion G**3** may be disposed adjacent to the third segmenting portion **403**. For example, the third ground portion G**3** may be disposed adjacent to an end, in the negative y-axis direction, of the second conductive portion **425** adjacent to the third segmenting portion **403**. In still another example, the third ground portion G**3** may be positioned between the third segmenting portion **403** and the third feeding point F**3**. Between the second feeding point F**2** and the third ground portion G**3**, the second ground portion G**2** and the third feeding point F**3** may be disposed. The second ground portion G**2** may be disposed between the second feeding point F**2** and the third feeding point F**3**. The third feeding point F**3** may be disposed between the second ground portion G**2** and the third ground portion G**3**.
(84) According to an embodiment, the second feeding point F**2** may be electrically connected to the wireless communication circuit **460** disposed on the PCB **340** through a second signal path **462**. A part of the second conductive portion **425** (e.g., a second radiator) may be electrically connected to the wireless communication circuit **460** through the second feeding point F**2**, thus operating as a second antenna.
(85) According to an embodiment, the second ground portion G**2** may be connected to a part of the support member **311**. The second ground portion G**2** may be electrically connected to the ground G formed on the PCB **340** through a part of the support member **311**. The second ground portion G**2** may ground a part (e.g., a first part) of the second conductive portion **425**.
(86) According to an embodiment, the third feeding point F**3** may be electrically connected to the wireless communication circuit **460** disposed on the PCB **340** through a third signal path **463**. At

least a part of the second conductive portion **425** (e.g., a second radiator) may be electrically connected to the wireless communication circuit **460** through the third feeding point F**3**, thus operating as a third antenna.

(87) According to an embodiment, the third ground portion G**3** may be connected, directly or indirectly, to a part of the support member **311**. The third ground portion G**3** may be electrically connected to the ground G formed on the PCB **340** through a part of the support member **311**. The third ground portion G**3** may ground another part (e.g., a second part) of the second conductive portion **425**.

(88) In an embodiment, FIG. **4**C is a diagram schematically illustrating an electrical connection relationship of the first conductive portion **415** and the second conductive portion **425**, shown in FIGS. **4**A and **4**B, with respect to the processor **450** and the wireless communication circuit **460**, disposed on the PCB **340**.

(89) With reference to FIG. **4**C, the PCB **340** may include the processor **450** (comprising processing circuitry), the wireless communication circuit **460**, the first matching circuit **451**, the second matching circuit **452**, and/or the ground G.

(90) According to an embodiment, the processor **450** (e.g., the processor **120** in FIG. **1**) may be electrically connected to the wireless communication circuit **460**, the first matching circuit **451**, and/or the second matching circuit **452**. In an embodiment, the processor **450** may control the wireless communication circuit **460**, the first matching circuit **451**, and/or the second matching circuit **452**. The processor **450** may control the wireless communication circuit **460** and thereby transmit a feeding signal to at least one of the first feeding point F**1**, the second feeding point F**2**, and the third feeding point F**3**. The processor **450** may control the first matching circuit **451** and/or the second matching circuit **452** and thereby control an electrical length or path of an antenna (e.g., the first to third antennas) including at least a part of the first conductive portion **415** and/or the second conductive portion **425**. The processor **450** may control on/off signals of the first matching circuit **451** and/or the second matching circuit **452** and may adjust and change the frequency band and resonant frequency of the first to third antennas using the first conductive portion **415** and the second conductive portion **425**. The processor **450** may include, for example, a communication processor, a radio frequency IC (RFIC), and/or a wireless communication module comprising communication circuitry (e.g., the wireless communication module **192** in FIG. **1**). Each "module" herein may comprise circuitry.

(91) According to an embodiment, the wireless communication circuit **460** may be electrically connected to the first feeding point F**1**, the second feeding point F**2**, and/or the third feeding point F**3**. For example, the wireless communication circuit **460** may be electrically connected to the first feeding point F**1** through the first signal path **461**. The wireless communication circuit **460** may be electrically connected to the second feeding point F**2** through the second signal path **462**. The wireless communication circuit **460** may be electrically connected to the third feeding point F**3** through the third signal path **463**. According to various embodiments, the wireless communication circuit **460** included may be at least one or more. For example, a plurality of wireless communication circuits **460** may be included and, depending on situations, electrically connected to the first feeding point F**1**, the second feeding point F**2**, and the third feeding point F**3**. In various embodiments, by using, for example, a contact pad, a coupling member, a C-clip, or a conductive foam spring, the wireless communication circuit **460** may be electrically connected to the first feeding point F**1**, the second feed point F**2**, and/or the third feeding point F**3**. The wireless communication circuit **460** may be electrically connected to the processor **450**. The wireless communication circuit **460** may transmit a feeding signal to the first feeding point F**1**, the second feeding point F**2**, and/or the third feeding point F**3**. The wireless communication circuit **460** may support the first conductive portion **415** and/or the second conductive portion **425** to transmit and/or receive radio signals.

(92) According to an embodiment, the first matching circuit **451** and/or the second matching circuit

452 may be electrically connected to the processor **450**. In another embodiment, the first matching circuit **451** and the second matching circuit **452** may be electrically connected to the wireless communication circuit **460**. The first matching circuit **451** may be electrically connected to the first point M**1** through the fourth signal path **454**. The second matching circuit **452** may be electrically connected to the second point M**2** through the fifth signal path **455**. The first matching circuit **451** and the second matching circuit **452** may include at least one switch or at least one lumped element. The at least one lumped element may include, for example, an inductor or a capacitor. The first matching circuit **451** and the second matching circuit **452** may perform an on/off operation under the control of the processor **450** or the wireless communication circuit **460**. As a matching value is adjusted under the control of the processor **450** or the wireless communication circuit **460**, the first matching circuit **451** and/or the second matching circuit **452** may adjust the electrical length or path of the antenna (e.g., the first to third antennas) including at least a part of the first conductive portion **415** and/or the second conductive portion **425**. For example, the first matching circuit **451** and/or the second matching circuit **452** may improve the radiation performance of the antenna including at least a part of the first conductive portion **415** and/or the second conductive portion **425**. The first matching circuit **451** and the second matching circuit **452** may electrically connect or disconnect the first point M**1** or the second point M**2** to or from the ground G under the control of the processor **450** or the wireless communication circuit **460**. The first matching circuit **451** and the second matching circuit **452** may be electrically connected to the ground G formed on the PCB **340**. In an embodiment, the first matching circuit **451** may include at least one inductor. In an embodiment, the second matching circuit **452** may include at least one capacitor. In another embodiment, the first matching circuit **451** may include at least one capacitor, and the second matching circuit **452** may include at least one inductor.

(93) According to an embodiment, the ground G formed on the PCB **340** may be connected to at least one of first ground portion G**1** disposed on the first conductive portion **415** and the second ground portion G**2** and the third ground portion G**3** disposed on the second conductive portion **425**.

(94) FIG. **5** is a diagram schematically illustrating the configurations of the first matching circuit and the second matching circuit shown in FIG. **4**C according to various example embodiments.

(95) With reference to FIG. **5**, each of the first matching circuit **451** or the second matching circuit **452** of the electronic device **300** according to various example embodiments may include at least one switch **510** and at least one passive element **520** (D**1**, D**2**, Dn, open) electrically connected to or disconnected from a signal path (e.g., the fourth signal path **454** and/or the fifth signal path **455**) by the at least one switch **510**. The at least one passive element **520** may have different element values. The at least one passive element **520** (e.g., a lumped element) may include a capacitor having various capacitance values and/or an inductor having various inductance values. The at least one switch **510** may selectively connect an element having a specified element value (e.g., a matching value) to the fourth signal path **454** and/or the fifth signal path **455** under the control of the processor **450**. In an embodiment, the at least one switch **510** may include a micro-electro mechanical systems (MEMS) switch. The MEMS switch performs a mechanical switching operation by an internal metal plate, thus having a complete turn on/off characteristic and not substantially affecting a change in antenna radiation characteristics. In some embodiments, the at least one switch **510** may include a single pole single throw (SPST), a single pole double throw (SPDT), or a switch having three or more throws.

(96) In an embodiment, the first matching circuit **451** and the first point M**1** may be electrically connected, directly or indirectly, using the fourth signal path **454**, and the second point M**2** may be electrically connected to the ground G. In another embodiment, the second matching circuit **452** and the second point M**2** may be electrically connected using the fifth signal path **455**, and the first point M**1** may be electrically connected to the ground G.

(97) FIG. **6**A is a diagram illustrating an operation principle of using, as a first antenna, a first conductive portion and a part of a second conductive portion of an electronic device according to

various example embodiments. FIG. **6**B is a diagram illustrating a flow of current applied to a first conductive portion and a part of a second conductive portion according to various example embodiments. FIG. **6**C is a diagram illustrating electric fields for a first conductive portion and a part of a second conductive portion according to various example embodiments.
(98) In an embodiment, FIG. **6**A is a diagram schematically showing configuration when the first conductive portion **415** and the second conductive portion **425** shown in FIGS. **4**A to **4**C according to various example embodiments are unfolded in a horizontal direction (e.g., the x-axis and negative x-axis directions).
(99) With reference to FIGS. **6**A to **6**C, the electronic device **300** according to an example embodiment may use the first conductive portion **415** and a part of the second conductive portion **425** as a radiator of a first antenna **610**. The first antenna **610** may operate in a first frequency band. For example, the first antenna **610** may transmit and/or receive a signal of about 600 MHz to 3 GHz.
(100) According to an embodiment, the first antenna **610** may support the first frequency band (e.g., about 600 MHz to 3 GHz) by using the first conductive portion **415** and a part of the second conductive portion **425** under the control of the processor **450** or the wireless communication circuit **460**. For example, the first antenna **610** may operate by using as a radiator the entirety of the first conductive portion **415** and a part of the second conductive portion **425** up to a placement point of the second ground portion G**2**.
(101) With reference to FIGS. **6**B and **6**C, in the first antenna **610**, a current flows from an end (e.g., in the negative y-axis direction, a point of the first conductive portion **415** where the second point M**2** is formed) of the first conductive portion **415** adjacent to the first segmenting portion **401** toward an end (e.g., in the y-axis direction, a point of the second conductive portion **425** where the second feeding point F**2** is formed) of the second conductive portion **425** adjacent to the first segmenting portion **401** through the first ground portion G**1**, a part of the support member **311** (e.g., the first support member **311** in FIG. **3**), and the second ground portion G**2** of the second conductive portion **425**, thereby forming an electric field (e.g., current distribution).
(102) According to an embodiment, in the case that the first antenna **610** operates in the first frequency band (e.g., about 600 MHz to 3 GHz) using the first conductive portion **415** and a part of the second conductive portion **425**, the first and second matching circuits **451** and **452** may be turned off, or the first matching circuit **451** may be turned on and the second matching circuit **452** may be turned off, under the control of the processor **450**. In an embodiment, in the case that the first antenna **610** operates at about 700 MHz to 750 MHz, the first and second matching circuits **451** and **452** may be turned off. In another embodiment, in the case that the first antenna **610** operates at about 800 MHz to 1 GHz, the first matching circuit **451** may be turned on, and the second matching circuit **452** may be turned off.
(103) According to an embodiment, as the first conductive portion **415** and a part of the second conductive portion **425** are used as a radiator, the electrical length of the first antenna **610** may be increased. In various embodiments, when the first antenna **610** operates in a frequency band of about 700 MHz to 750 MHz and/or a frequency band of about 800 MHz to 1 GHz, a second antenna (e.g., the second antenna **720** in FIGS. **7**A to **7**C) does not operate, and a third antenna (e.g., the third antenna **830** in FIG. **8**A) may operate by receiving a feeding signal through the third feeding point F**3**.
(104) FIG. **7**A is a diagram illustrating an operation principle of using, as a second antenna, an end of a first conductive portion and a part of a second conductive portion of an electronic device according to various example embodiments. FIG. **7**B is a diagram illustrating a flow of current applied to a part of a first conductive portion and a part of a second conductive portion according to various example embodiments. FIG. **7**C is a diagram illustrating electric fields for a part of a first conductive portion and a part of a second conductive portion according to various example embodiments.

(105) In an embodiment, FIG. **7**A is a diagram schematically showing configuration when the first conductive portion **415** and the second conductive portion **425** shown in FIGS. **4**A to **4**C according to various example embodiments are unfolded in a horizontal direction (e.g., the x-axis and negative x-axis directions).
(106) With reference to FIGS. **7**A to **7**C, the electronic device **300** according to an example embodiment may use an end (e.g., in the negative y-axis direction, a point of the first conductive portion **415** where the second point M**2** is formed) of the first conductive portion **415** and a part of the second conductive portion **425** as a radiator of a second antenna **720**. The second antenna **720** may operate in a second frequency band. For example, the second antenna **720** may transmit and/or receive a signal of about 3 GHz to 4.5 GHz.
(107) According to an embodiment, the second antenna **720** may support the second frequency band (e.g., about 3 GHz to 4.5 GHz) by using an end (e.g., in the negative y-axis direction, a point of the first conductive portion **415** where the second point M**2** is formed) of the first conductive portion **415** and a part of the second conductive portion **425** under the control of the processor **450** or the wireless communication circuit **460**. For example, the second antenna **720** may operate by using as a radiator the end of the first conductive portion **415** and a part of the second conductive portion **425** up to a placement point of the second ground portion G**2**.
(108) With reference to FIGS. **7**B and **7**C, in the second antenna **720**, a current flows from an end (e.g., in the y-axis direction, a point of the second conductive portion **425** where the second feeding point F**2** is formed) of the second conductive portion **425** toward an end (e.g., in the negative y-axis direction, a point of the first conductive portion **415** where the second point M**2** is formed) of the first conductive portion **415** adjacent to the first segmenting portion **401** through the second ground portion G**2** and a part of the support member **311**, thereby forming an electric field (e.g., current distribution).
(109) According to an embodiment, in the case that the second antenna **720** operates in the second frequency band (e.g., about 3 GHz to 4.5 GHz) using a part of the first conductive portion **415** and a part of the second conductive portion **425**, the first matching circuit **451** may be turned off and the second matching circuit **452** may be turned on, or the first and second matching circuits **451** and **452** may be turned on, under the control of the processor **450**. In an embodiment, in the case that the first matching circuit **451** is turned off and the second matching circuit **452** is turned on, the first antenna **610** may not operate, the second antenna **720** may operate in the second frequency band, and a third antenna (e.g., the third antenna **830** in FIG. **8**A) may operate by receiving a feeding signal through the third feeding point F**3**. In another embodiment, in the case that the first and second matching circuits **451** and **452** are turned on, at least a part of the first conductive portion **415** may be used as the first antenna **610** operating at about 1.5 GHz to 3 GHz, the second antenna **720** may operation in the second frequency band, and the third antenna (e.g., the third antenna **830** in FIG. **8**A) may operate by receiving a feeding signal through the third feeding point F**3**.
(110) According to various embodiments, in the case that the first matching circuit **451** is turned on, the second matching circuit **452** is switched to the on state, the matching value of the first matching circuit **451** and/or the second matching circuit **452** is changed, and thereby at least a part of the first conductive portion **415** is used as the first antenna **610**, the first antenna **610** shown in FIG. **7**B may be controlled to operate at about 1.5 GHz to 3 GHz, compared to the frequency band of about 700 MHz to 750 MHz shown in FIG. **6**B.
(111) According to an embodiment, as a part of the first conductive portion **415** and a part of the second conductive portion **425** share the first segmenting portion **401** and are used as an antenna radiator, it is possible to improve isolation in which the antenna can be separated through the segmenting portion formed in the housing **310**.
(112) FIG. **8**A is a diagram illustrating an operation principle of using, as a third antenna, a part of a first conductive portion and at least a part of a second conductive portion of an electronic device

according to various example embodiments. FIG. **8**B is a diagram illustrating a flow of current applied to a part of a first conductive portion and at least a part of a second conductive portion according to various example embodiments. FIG. **8**C is a diagram illustrating a radiation pattern of a third antenna of an electronic device according to various example embodiments. FIG. **8**D is a diagram illustrating radiation efficiency for a third antenna of an electronic device according to various example embodiments and an antenna of an electronic device according to a comparative embodiment.

(113) In an embodiment, FIG. **8**A is a diagram schematically showing configuration when the first conductive portion **415** and the second conductive portion **425** shown in FIGS. **4**A to **4**C according to various example embodiments are unfolded in a horizontal direction (e.g., the x-axis and negative x-axis directions).

(114) In various embodiments, the second conductive portion **425** shown in FIG. **8**A may be configured to include an extension part **825** extending in a direction (e.g., in the y-axis direction) of the first segmenting portion **401** from a point where the second feeding point F**2** is formed, compared to FIGS. **4**A and **4**B. For example, the first segmenting portion **401** shown in FIGS. **4**A and **4**B is formed adjacent to the point where the second feeding point F**2** is formed, but the first segmenting portion **401** shown in FIG. **8**A may be formed at a location further away, as much as the extension part **825**, from the point where the second feeding point F**2** is formed.

(115) With reference to FIGS. **8**A and **8**B, the electronic device **300** according to an example embodiment may use an end of the first conductive portion **415** and at least a part of the second conductive portion **425** as a radiator of a third antenna **830**. The third antenna **830** may operate in a third frequency band. For example, the third antenna **830** may transmit and/or receive a signal of about 6.5 GHz to 8 GHz.

(116) According to an embodiment, the third antenna **830** may support the third frequency band (e.g., about 6.5 GHz to 8 GHz) by using an end (e.g., in the negative y-axis direction, a point of the first conductive portion **415** where the second point M**2** is formed) of the first conductive portion **415** and at least a part of the second conductive portion **425** under the control of the processor **450** or the wireless communication circuit **460**. For example, the third antenna **830** may operate by using as a radiator the end (e.g., in the negative y-axis direction, a point of the first conductive portion **415** where the second point M**2** is formed) of the first conductive portion **415** and a part of the second conductive portion **425** up to a placement point of the third ground portion G**3**.

(117) With reference to FIG. **8**B, in the third antenna **830**, an electric field (e.g., current distribution) may be formed in a part of the first conductive portion **415** and at least a part of the second conductive portion **425**. With reference to FIG. **8**C, the third antenna **830** may form a radiation pattern having omnidirectional directivity with respect to the electronic device **300**.

(118) The electronic device **300** according to an example embodiment enables the first conductive portion **415** and the second conductive portion **425** disposed on the housing **310** to selectively share the first segmenting portion **401**, and uses them as an antenna, thereby improving radiation performance (e.g., directivity) compared to a laser direct structuring (LDS) antenna according to a comparative embodiment disposed inside the housing **310**.

(119) With reference to FIG. **8**D, it can be seen that the radiation efficiency S**1** of the electronic device **300** according to an example embodiment, in which a part of the first conductive portion **415** and at least a part of the second conductive portion **425** sharing one segmenting portion (e.g., the first segmenting portion **401**) are used as the third antenna **830**, is improved, for example, in a frequency band of about 6250 MHz to 6650 MHz and a frequency band of about 7600 MHz to 8400 MHz, compared to the radiation efficiency C**1** of the electronic device according to the comparative embodiment that uses an opening formed between the housing and the display without sharing a segmenting portion because there is no segmenting portion.

(120) FIG. **9** is a diagram schematically illustrating various configurations of a first conductive portion and a second conductive portion of an electronic device according to various example

embodiments.
(121) According to an embodiment, in the electronic device **300** shown in FIGS. **4**A to **4**C, the first conductive portion **415** may be disposed between the first segmenting portion **401** formed on the first side surface **410** and the second segmenting portions **402** formed on the second side surface **420**, and the second conductive portion **425** may be disposed between the first segmenting portion **401** and the third segmenting portion **403** formed on the first side surface **410**. According to various embodiments, the first ground portion G**1**, the first feeding point F**1**, the first point M**1**, and/or the second point M**2** may be disposed inwardly from the second conductive portion **425**, and also the second feeding point F**2**, the second ground portion G**2**, the third feeding point F**3**, and/or the third ground portion G**3** may be disposed inwardly from the first conductive portion **415**.
(122) The electronic device **300** according to various example embodiments is not limited to the aforesaid positions and number of the first segmenting portion **401**, the second segmenting portion **402**, the third segmenting portion **403**, the first conductive portion **415**, and the second conductive portion **425**, and various other embodiments may be possible as long as a plurality of antennas can share and use one segmenting portion (e.g., the first segmenting portion **401**).
(123) According to various embodiments, in the electronic device **300** according to various example embodiments, the first conductive portion **415** may be disposed on the first side surface **410**, and the second conductive portion **425** may be disposed on the second side surface **420** and/or the third side surface **430**. In the case that the first conductive portion **415** is disposed on the first side surface **410**, the first conductive portion **415** may have inwardly the second feeding point F**2**, the second ground portion G**2**, the third feeding point F**3**, and/or the third ground portion G**3**. In the case that the second conductive portion **425** is disposed on the second side surface **420** and/or the third side surface **430**, the second conductive portion **425** may have inwardly the first ground portion G**1**, the first feeding point F**1**, the first point M**1**, and/or the second point M**2**. In various embodiments, the positions and number of the first ground portion G**1**, the first feeding point F**1**, the first point M**1**, the second point M**2**, the second feeding point F**2**, the second ground portion G**2**, the third feeding point F**3**, and the third ground portion G**3** are not limited to the above-described embodiment and may be variously changed and modified.
(124) According to an example embodiment, an electronic device **101**, **200**, or **300** may include a housing **310** including a first side surface **410** and a second side surface **420**; a support member **311** disposed inside the housing **310** and connected to a part of the first and second side surfaces **410** and **420**; a first opening **406** formed between the first side surface **410** and the support member **311**, and a second opening **408** formed among a part of the first side surface **410**, the second side surface **420**, and the support member **311**; a printed circuit board **340** disposed on, directly or indirectly, the support member **311** and having a ground G; a first conductive portion **415** disposed between a first segmenting portion **401** formed in the first side surface **410** and a second segmenting portion **402** formed in the second side surface **420**, and including a first ground portion G**1**, a first feeding point F**1**, a first point M**1**, and/or a second point M**2**; a second conductive portion **425** disposed between the first segmenting portion **401** and a third segmenting portion **403** formed in the first side surface **410**, and including a second feeding point F**2**, a second ground portion G**2**, a third feeding point F**3**, and/or a third ground portion G**3**; a wireless communication circuit **460** electrically connected to the first feeding point F**1**, the second feeding point F**2**, and/or the third feeding point F**3**; a processor **450** electrically connected to the wireless communication circuit **460**; and a first matching circuit **451** electrically connected to the ground G and the first point M**1** and/or a second matching circuit **452** electrically connected to the ground G and the second point M**2**, wherein the first conductive portion **415** and at least a part of the second conductive portion **425** may be configured to operate as at least one antenna, depending on an operation of the first matching circuit **451** or the second matching circuit **452** corresponding to control of the processor **450**.
(125) According to an embodiment, the first conductive portion **415** and the at least a part of the

second conductive portion **425** may be configured to operate as a plurality of antennas by sharing and using the first segmenting portion **401**.
(126) According to an embodiment, the first conductive portion **415** may be disposed in a "¬" shape between the first segmenting portion **401** and the second segmenting portion **402**.
(127) According to an embodiment, the first ground portion G**1** may be disposed between the second segmenting portion **402** and the first feeding point F**1**, the second point M**2** may be disposed between the first segmenting portion **401** and the first point M**1**, and the first point M**1** may be disposed between the first feeding point F**1** and the second point M**2**.
(128) According to an embodiment, the second feeding point F**2** may be disposed between the first segmenting portion **401** and the second ground portion G**2**, the third ground portion G**3** may be disposed between the third segmenting portion **403** and the second feeding point F**2**, and the third feeding point F**3** may be disposed between the second ground portion G**2** and the third ground portion G**3**.
(129) According to an embodiment, the processor **450** may be configured to adjust a matching value of the first matching circuit **451** and/or the second matching circuit **452**, and to adjust an electrical length or path of an antenna including the first conductive portion **415** and/or the at least a part of the second conductive portion **425**.
(130) According to an embodiment, the first matching circuit **451** may include at least one inductor, and the second matching circuit **452** may include at least one capacitor.
(131) According to an embodiment, the first conductive portion **415** electrically connected to the first and second matching circuits **451** and **452** and a part included in the second conductive portion **425** and corresponding to the second ground portion G**2** may be configured to operate as a first antenna **610**.
(132) According to an embodiment, in a case that the first antenna **610** operates in a first frequency band, the processor **450** may control the first and second matching circuits **451** and **452** to be turned off, or control the first matching circuit **451** to be turned on and the second matching circuit **452** to be turned off.
(133) According to an embodiment, in a case that the first and second matching circuits **451** and **452** are turned off, the first antenna **610** may be configured to operate in a frequency band of 700 MHz to 750 MHz.
(134) According to an embodiment, in a case that the first matching circuit **451** is turned on and the second matching circuit **452** is turned off, the first antenna **610** may be configured to operate in a frequency band of 800 MHz to 1 GHz.
(135) According to an embodiment, a part from the second ground portion G**2** included in the second conductive portion **425** to the second point M**2** included in the first conductive portion **415** and electrically connected to the second matching circuit **452** may be configured to operate as a second antenna **720**.
(136) According to an embodiment, in a case that the second antenna **720** operates in a second frequency band, the processor **450** may control the first matching circuit **451** to be turned off and the second matching circuit **452** to be turned on, or control the first and second matching circuits **451** and **452** to be turned on.
(137) According to an embodiment, a part from the second point M**2** included in the first conductive portion **415** and electrically connected to the second matching circuit **452** to the third ground portion G**3** included in the second conductive portion **425** may be configured to operate as a third antenna **830** supporting a third frequency band.
(138) According to an embodiment, the third antenna **830** may be configured to operate in a state where the first matching circuit **451** is turned on and/or off and the second matching circuit **452** is turned on and/or off.
(139) According to an example embodiment, an electronic device **101**, **200**, or **300** may include a housing **310** including a first side surface **410** and a second side surface **420**; a support member **311**

disposed inside the housing **310** and connected, directly or indirectly, to a part of the first and second side surfaces **410** and **420**; a first opening **406** formed between the first side surface **410** and the support member **311**, and a second opening **408** formed among a part of the first side surface **410**, the second side surface **420**, and the support member **311**; a printed circuit board **340** disposed on, directly or indirectly, the support member **311** and having a ground G; a first conductive portion **415** disposed between a first segmenting portion **401** formed in the first side surface **410** and a second segmenting portion **402** formed in the second side surface **420**, and including a first ground portion G**1**, a first feeding point F**1**, a first point M**1**, and/or a second point M**2**; a second conductive portion **425** disposed between the first segmenting portion **401** and a third segmenting portion **403** formed in the first side surface **410**, and including a second feeding point F**2**, a second ground portion G**2**, a third feeding point F**3**, and/or a third ground portion G**3**; a wireless communication circuit **460** electrically connected, directly or indirectly, to the first feeding point F**1**, the second feeding point F**2**, and/or the third feeding point F**3**; a processor **450** electrically connected, directly or indirectly, to the wireless communication circuit **460**; and a first matching circuit **451** electrically connected, directly or indirectly, to the ground G and the first point M**1** and/or a second matching circuit **452** electrically connected, directly or indirectly, to the ground G and the second point M**2**, wherein depending on an operation of the first matching circuit **451** or the second matching circuit **452** corresponding to control of the processor **450**, the first conductive portion **415** electrically connected, directly or indirectly, to the first and second matching circuits **451** and **452** and a part included in the second conductive portion **425** and corresponding to the second ground portion G**2** may be configured to operate as a first antenna **610**, and wherein a part from the second ground portion G**2** included in the second conductive portion **425** to the second point M**2** included in the first conductive portion **415** and electrically connected, directly or indirectly, to the second matching circuit **452** may be configured to operate as a second antenna **720**.

(140) According to an embodiment, in a case that the first antenna **610** operates in a first frequency band, the processor **450** may control the first and second matching circuits **451** and **452** to be turned off, and/or control the first matching circuit **451** to be turned on and the second matching circuit **452** to be turned off.

(141) According to an embodiment, in a case that the first and second matching circuits **451** and **452** are turned off, the first antenna **610** may be configured to operate in a frequency band of 700 MHz to 750 MHz.

(142) Each embodiment herein may be used in combination with any other embodiment(s) described herein.

(143) According to an embodiment, in a case that the first matching circuit **451** is turned on and the second matching circuit **452** is turned off, the first antenna **610** may be configured to operate in a frequency band of 800 MHz to 1 GHz.

(144) According to an embodiment, in a case that the second antenna **720** operates in a second frequency band, the processor **450** may control the first matching circuit **451** to be turned off and the second matching circuit **452** to be turned on, or control the first and second matching circuits **451** and **452** to be turned on.

(145) Hereinbefore, although the disclosure has been described using a variety of embodiments, it is natural that all changes and modifications made by those of ordinary skill in the art to which the disclosure pertains without departing from the technical scope of the disclosure also belong to the disclosure. While the disclosure has been illustrated and described with reference to various embodiments, it will be understood that the various embodiments are intended to be illustrative, not limiting. It will further be understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. An electronic device comprising: a housing including a first side surface and a second side surface; a support member, comprising a support, disposed at least partially inside the housing and connected to a part of the first and second side surfaces; a first opening between at least the first side surface and the support member, and a second opening provided among a part of the first side surface, the second side surface, and the support member; a printed circuit board disposed on the support member and comprising a ground; a first conductive portion, comprising conductive material, disposed between at least a first segmenting portion provided in the first side surface and a second segmenting portion provided in the second side surface, and including at least one of: a first ground portion, a first feeding point, a first point, and/or a second point; a second conductive portion, comprising conductive material, disposed between at least the first segmenting portion and a third segmenting portion provided in the first side surface, and including at least one of: a second feeding point, a second ground portion, a third feeding point, and/or a third ground portion; a wireless communication circuit electrically connected to at least one of the first feeding point, the second feeding point, and/or the third feeding point; a processor electrically connected to the wireless communication circuit; and a first matching circuit electrically connected to the ground and the first point and/or a second matching circuit electrically connected to the ground and the second point, wherein the first conductive portion and at least a part of the second conductive portion are configured to operate as at least one antenna, based on an operation of the first matching circuit and/or the second matching circuit based on control of the processor.
2. The electronic device of claim 1, wherein the first conductive portion and the at least a part of the second conductive portion are configured to operate as a plurality of antennas at least by sharing and using the first segmenting portion.
3. The electronic device of claim 1, wherein the first conductive portion is disposed in a “¬” shape between at least the first segmenting portion and the second segmenting portion.
4. The electronic device of claim 1, wherein the first ground portion is disposed between at least the second segmenting portion and the first feeding point, the second point is disposed between at least the first segmenting portion and the first point, and the first point is disposed between at least the first feeding point and the second point.
5. The electronic device of claim 1, wherein the second feeding point is disposed between at least the first segmenting portion and the second ground portion, the third ground portion is disposed between at least the third segmenting portion and the second feeding point, and the third feeding point is disposed between at least the second ground portion and the third ground portion.
6. The electronic device of claim 1, wherein the processor is configured to adjust a matching value of the first matching circuit and/or the second matching circuit, and to adjust an electrical length or path of an antenna including the first conductive portion and/or the at least a part of the second conductive portion.
7. The electronic device of claim 1, wherein the first matching circuit includes at least one inductor, and the second matching circuit includes at least one capacitor.
8. The electronic device of claim 1, wherein the first conductive portion is electrically connected to the first and second matching circuits and a part included in the second conductive portion and corresponding to the second ground portion are configured to operate as a first antenna.
9. The electronic device of claim 8, wherein based on operation of the first antenna in a first frequency band, the processor is configured to control the first and second matching circuits to be turned off, or to control the first matching circuit to be turned on and the second matching circuit to be turned off.
10. The electronic device of claim 9, wherein based on the first and second matching circuits being turned off, the first antenna is configured to operate in a frequency band of 700 MHz to 750 MHz.

11. The electronic device of claim 9, wherein based on the first matching circuit being on and the second matching circuit being off, the first antenna is configured to operate in a frequency band of 800 MHz to 1 GHz.
12. The electronic device of claim 8, wherein a part from the second ground portion included in the second conductive portion to the second point included in the first conductive portion and electrically connected to the second matching circuit is configured to operate as a second antenna.
13. The electronic device of claim 12, wherein based on operation of the second antenna in a second frequency band, the processor is configured to control the first matching circuit to be turned off and the second matching circuit to be turned on, or to control the first and second matching circuits to be turned on.
14. The electronic device of claim 1, wherein a part from the second point included in the first conductive portion and electrically connected to the second matching circuit to the third ground portion included in the second conductive portion is configured to operate as a third antenna supporting a third frequency band.
15. The electronic device of claim 14, wherein the third antenna is configured to operate at least in a state where the first matching circuit is turned on and/or off and the second matching circuit is turned on and/or off.
16. An electronic device comprising: a housing including a first side surface and a second side surface; a support member disposed at least partially inside the housing and connected to a part of the first and second side surfaces; a first opening between at least the first side surface and the support member, and a second opening provided in at least one of: a part of the first side surface, the second side surface, and the support member; a printed circuit board disposed on the support member and comprising a ground; a first conductive portion, comprising conductive material, disposed between at least a first segmenting portion provided at least partially in the first side surface and a second segmenting portion provided at least partially in the second side surface, and including at least one of: a first ground portion, a first feeding point, a first point, and/or a second point; a second conductive portion, comprising conductive material, disposed between at least the first segmenting portion and a third segmenting portion provided at least partially in the first side surface, and including at least one of: a second feeding point, a second ground portion, a third feeding point, and/or a third ground portion; a wireless communication circuit electrically connected to at least one of: the first feeding point, the second feeding point, and/or the third feeding point; a processor electrically connected to the wireless communication circuit; and a first matching circuit electrically connected to the ground and the first point and/or a second matching circuit electrically connected to the ground and the second point, wherein based on an operation of the first matching circuit and/or the second matching circuit corresponding to control of the processor, the first conductive portion electrically connected to the first and second matching circuits and a part included in the second conductive portion and corresponding to the second ground portion are configured to operate as a first antenna, and wherein a part from the second ground portion included in the second conductive portion to the second point included in the first conductive portion and electrically connected to the second matching circuit is configured to operate as a second antenna.
17. The electronic device of claim 16, wherein based on operation of the first antenna in a first frequency band, the processor is configured to control the first and second matching circuits to be turned off, and/or to control the first matching circuit to be turned on and the second matching circuit to be turned off.
18. The electronic device of claim 17, wherein based on the first and second matching circuits being turned off, the first antenna is configured to operate in a frequency band of 700 MHz to 750 MHz.
19. The electronic device of claim 17, wherein based on the first matching circuit being turned on and the second matching circuit being turned off, the first antenna is configured to operate in a

frequency band of 800 MHz to 1 GHz.

20. The electronic device of claim 16, wherein based on operation of the second antenna in a second frequency band, the processor is configured to control the first matching circuit to be turned off and the second matching circuit to be turned on, and/or to control the first and second matching circuits to be turned on.